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SCHOOL OF ENGINEERING

A PROJECT REPORT ON

“EARTHQUAKE MAGNITUDE PREDICTION & RISK CLASSIFICATION”

Submitted By

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DECLARATION

I, **SAI KRISHA K**, hereby declare that this project work entitled **EARTHQUAKE MAGNITUDE PREDICTION & RISK CLASSIFICATION** is submitted in partial fulfilment for the award of the degree of **BCA** of **Chanakya University**.
I further declare that I have not submitted this project report either in part or in full to any other university for the award of any degree.

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ABSTRACT

Earthquakes are one of the most destructive natural phenomena, posing significant threats to human life and infrastructure. Accurate prediction and classification of earthquake risk levels support the development of early warning systems and enhance disaster preparedness. This study introduces a comprehensive data mining methodology to predict earthquake magnitude and classify seismic events by risk level using real-world earthquake data.

The dataset comprises global earthquake records with key attributes including latitude, longitude, depth, and magnitude. Nine features are engineered from the raw data, such as logarithmic depth transformation and cyclic trigonometric encodings of geographic coordinates, to more effectively capture spatial patterns. Risk labels (Low, Medium, High) are assigned based on magnitude thresholds informed by seismological expertise.

In order to forecast earthquake magnitude, two regression models—Linear Regression and Random Forest Regressor—are trained and assessed using MAE, RMSE, and R^2 using five-fold cross-validation. In order to explicitly minimize data leaking, two classification models—Decision Tree and Random Forest Classifier—are trained to categorize earthquake risk level by removing magnitude from classifier inputs.

The outcomes demonstrate that Random Forest performs better than Linear Regression in both tasks. The project uses an interactive Folium map to visualize the distribution of global risk and further aggregates projections by nation. Important results show that significant risk patterns do appear across geographic zones, even if magnitude cannot be accurately anticipated from location and depth alone.

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INTRODUCTION

1.1 Introduction

Tectonic plate movement beneath the Earth's surface is what causes earthquakes, which are naturally occurring geological phenomena. They can cause tsunamis, demolish infrastructure, and kill thousands of people in a matter of seconds, making them some of the most destructive natural disasters. Because seismic processes are inherently complicated and variable, accurately predicting earthquake magnitude remains a difficult task despite advancements in seismology.

Data mining techniques offer a powerful alternative approach to earthquake analysis. Rather than attempting deterministic physical prediction, machine learning models can identify hidden patterns in historical earthquake data and produce useful predictions or classifications. This project applies data mining and machine learning to a real-world global earthquake dataset to address two primary tasks: magnitude prediction (regression) and risk level classification.

The project is implemented entirely in Python using Google Colab, leveraging libraries such as scikit-learn, pandas, matplotlib, seaborn, and folium. The results are presented through multiple visualizations including scatter plots, heatmaps, decision tree diagrams, feature importance charts, and an interactive global risk map.

1.2 Objectives

- To preprocess and engineer features from a real-world earthquake dataset.
- To predict earthquake magnitude using Linear Regression and Random Forest Regressor.
- To classify earthquake events into risk levels (Low, Medium, High) using Decision Tree and Random Forest Classifier.
- To evaluate all models using appropriate metrics: MAE, RMSE, R^2 for regression; Accuracy, Precision, Recall for classification.
- To validate models using 5-fold cross-validation

- To visualize results through correlation heatmaps, feature importance charts, and an interactive Folium map.
- To aggregate predicted risk by country and generate a country-level risk summary.

1.3 Scope of the Project

This project covers the full data mining pipeline from raw data ingestion to model evaluation and visualization. It handles global earthquake records with attributes including latitude, longitude, depth, magnitude, and place name. The scope includes feature engineering, model training, comparative evaluation, and geospatial visualization. The project does not aim to build a real-time prediction system but serves as a comprehensive analytical study demonstrating the applicability of data mining to seismic risk assessment.

CHAPTER-2

PROBLEM STATEMENT

2.1 Problem Statement

Earthquake events are logged globally by agencies such as the United States Geological Survey (USGS) with attributes including location (latitude, longitude), depth, and magnitude. Despite the availability of large historical datasets, several analytical challenges remain:

- Can earthquake magnitude be predicted from purely geographic and depth-based features using machine learning?
- Can earthquake events be accurately classified into risk categories (Low, Medium, High) without using magnitude as a model input, thereby avoiding data leakage?
- Which geographic regions and countries are most prone to high-risk seismic activity?
- Which features contribute most to predicting seismic outcomes?

The main scientific and business issue this project attempts to solve is how effectively machine learning models can classify the danger level and estimate the magnitude of a seismic event given just its depth and geographic position (latitude and longitude). This is pertinent to government organizations in charge of infrastructure planning in earthquake-prone areas, disaster preparedness organizations, and civil engineers.

Making sure there is no data leakage in the categorization model is a secondary difficulty. Including magnitude as a feature would easily solve the classification problem because risk labels are derived from magnitude. In order to make the task really predictive, this project purposefully removes magnitude from classifier inputs and solely uses spatial and depth-derived variables.

CHAPTER-3

METHODOLOGY

3.1 Data Overview

The dataset used in this project is past 30 days global earthquake CSV file (earthquake.csv) sourced from USGS earthquake records. It contains thousands of seismic event entries with the following primary columns:

Column	Type	Description
latitude	Float	Geographic latitude of the earthquake
longitude	Float	Geographic longitude of the earthquake
depth	Float	Depth of earthquake focus in km
mag	Float	Magnitude of the earthquake (Richter scale)
place	String	USGS place description (e.g., “15km NE of Tokyo, Japan”)

Dataset link: https://earthquake.usgs.gov/earthquakes/feed/v1.0/csv.php
Data chosen from the source was past 30 days (updated every minute) of M2.5+ Earthquakes.

Data preprocessing involves the following steps:

- Column name normalization: All column names are stripped, lowercased, and mapped to standardized names using a rename_map dictionary.
- Null value removal: Rows with missing values in required columns (latitude, longitude, depth, mag) are dropped.
- Outlier filtering: Records with depth < 0, depth > 800 km, or magnitude ≤ 0 are removed as physically invalid.
- Country extraction: The country or region name is extracted from the USGS place field by splitting on the comma separator and taking the last component.

After cleaning, the following feature engineering is applied to create 9 model input features:

Feature	Description
latitude	Raw latitude coordinate

longitude	Raw longitude coordinate
depth	Raw depth in km
depth_log	Log-transformed depth: log(1 + depth) to reduce right skew
lat_abs	Absolute value of latitude (distance from equator)
lon_sin / lon_cos	Cyclic trigonometric encoding of longitude
lat_sin / lat_cos	Cyclic trigonometric encoding of latitude

Rows before cleaning: 2061
Rows after cleaning: 2058

Risk labels are assigned to each event based on magnitude thresholds derived from seismological standards:

Risk Label	Magnitude Range	Class Code
Low Risk	mag < 4.5	0
Medium Risk	4.5 ≤ mag ≤ 6.0	1
High Risk	mag > 6.0	2

3.2 Key Metrics and KPIs

The project applies two types of machine learning tasks, each evaluated using appropriate metrics.

Regression Metrics (for Magnitude Prediction):

- Mean Absolute Error (MAE): Average absolute difference between actual and predicted magnitude.
- Root Mean Squared Error (RMSE): Square root of average squared differences. More sensitive to large errors.
- R^2 Score: Proportion of variance in magnitude explained by the model. Range: 0 to 1 (higher is better).
- 5-Fold Cross-Validation R^2: Average R^2 across 5 data splits to ensure generalizability.

Classification Metrics (for Risk Level Prediction):

- Accuracy: Proportion of correctly classified risk labels.

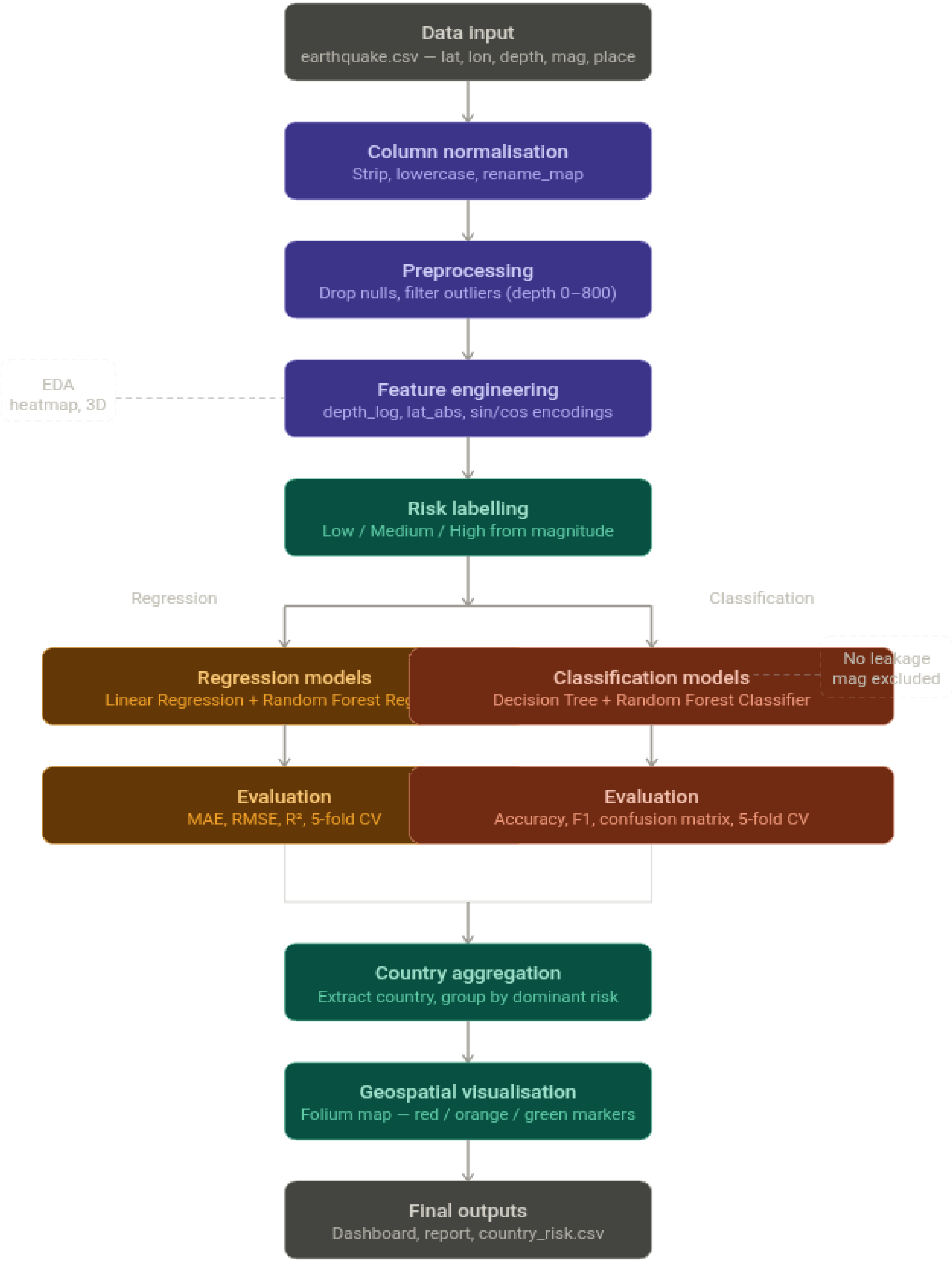
- Precision: Of all predicted positives, how many are actually positive.
- Recall: Of all actual positives, how many were correctly identified.
- F1-Score: Harmonic mean of precision and recall.
- Confusion Matrix: Table showing True Positives, False Positives, True Negatives, False Negatives for each class.
- 5-Fold Cross-Validation Accuracy: Average accuracy across 5 stratified splits.

Models Applied:

- Linear Regression — Baseline regression model using scaled features (StandardScaler applied).
- Random Forest Regressor — Ensemble of 100 decision trees for magnitude prediction. No scaling required.
- Decision Tree Classifier — Single tree with max_depth=5 for risk classification.
- Random Forest Classifier — Ensemble of 100 trees for risk classification with stratified train-test split (80/20).

For all models, a train-test split of 80% training and 20% testing is applied with random_state=42 for reproducibility. Classification uses stratified splitting to maintain class proportions across splits.

FLOWCHART OF THE MODEL


Data input
earthquake.csv — lat, lon, depth, mag, place
Column normalisation
Strip, lowercase, rename_map
Preprocessing
Drop nulls, filter outliers (depth 0–800)
EDA
heatmap, 3D
Feature engineering
depth_log, lat_abs, sin/cos encodings
Risk labelling
Low / Medium / High from magnitude
Regression
Classification
Regression models
Linear Regression + Random Forest Reg
Classification models
Decision Tree + Random Forest Classifier
No leakage
mag excluded
Evaluation
MAE, RMSE, R², 5-fold CV
Evaluation
Accuracy, F1, confusion matrix, 5-fold CV
Country aggregation
Extract country, group by dominant risk
Geospatial visualisation
Folium map — red / orange / green markers
Final outputs
Dashboard, report, country_risk.csv

CHAPTER-4
INSIGHTS AND ANALYSIS

4.1 Insights

Exploratory Data Analysis (EDA)

The correlation heatmap (Figure 4.1) reveals that geographical location features, especially latitude and longitude, show stronger relationships with earthquake magnitude than depth-related features. Latitude shows a moderate negative correlation (-0.60), while longitude shows a moderate positive correlation (0.63) with magnitude. The derived feature lat_abs also shows a negative correlation (-0.50). In contrast, depth and depth_log show only weak positive correlations, indicating that earthquake magnitude depends more on spatial location than depth alone. This supports the idea that tectonic plate boundaries and regional seismic zones are major drivers of earthquake activity.

The 3D scatter plot (Figure 4.2) visualizes earthquake magnitude as a function of longitude and latitude, confirming that high-magnitude events are geographically clustered along known tectonic plate boundaries, particularly around the Pacific Ring of Fire.

The risk distribution chart (Figure 4.3) illustrates the dataset's class imbalance: most occurrences are categorized as Low Risk, followed by Medium Risk, and High-Risk events are very uncommon. The categorization models take this imbalance into account.

Risk Label Distribution:

Risk Label	Count
Low Risk	1495 (72.6%)
Medium Risk	557 (27.1%)
High Risk	6 (0.3%)

Figure 4.1: *Feature Correlation Heatmap*

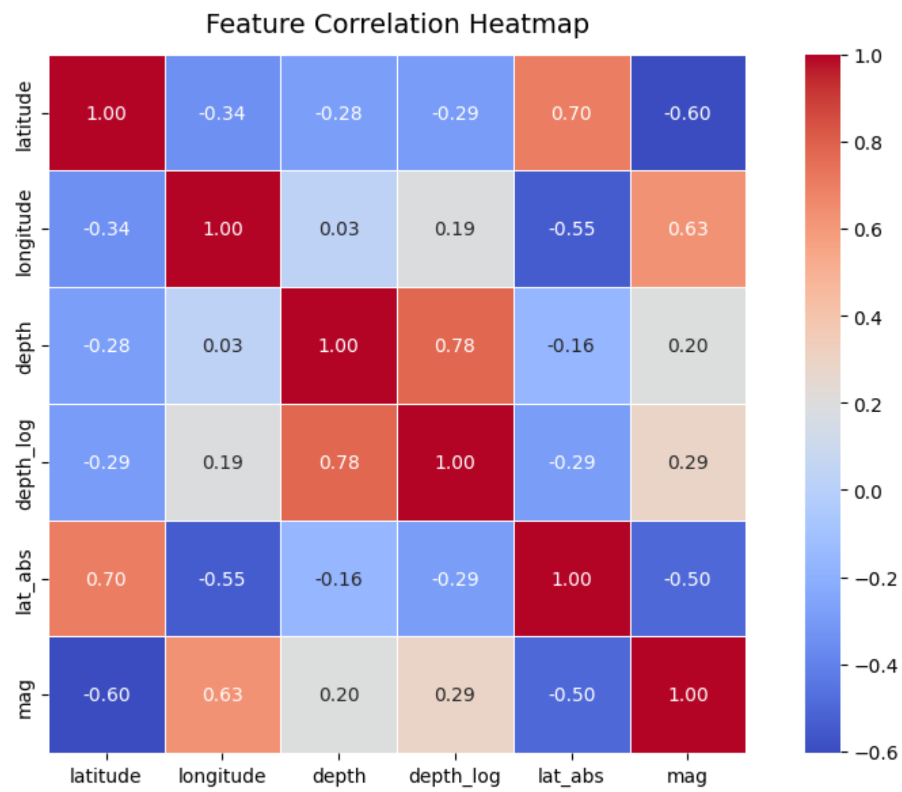


Figure 4.2: *3D Spatial Distribution of Earthquake Magnitudes*

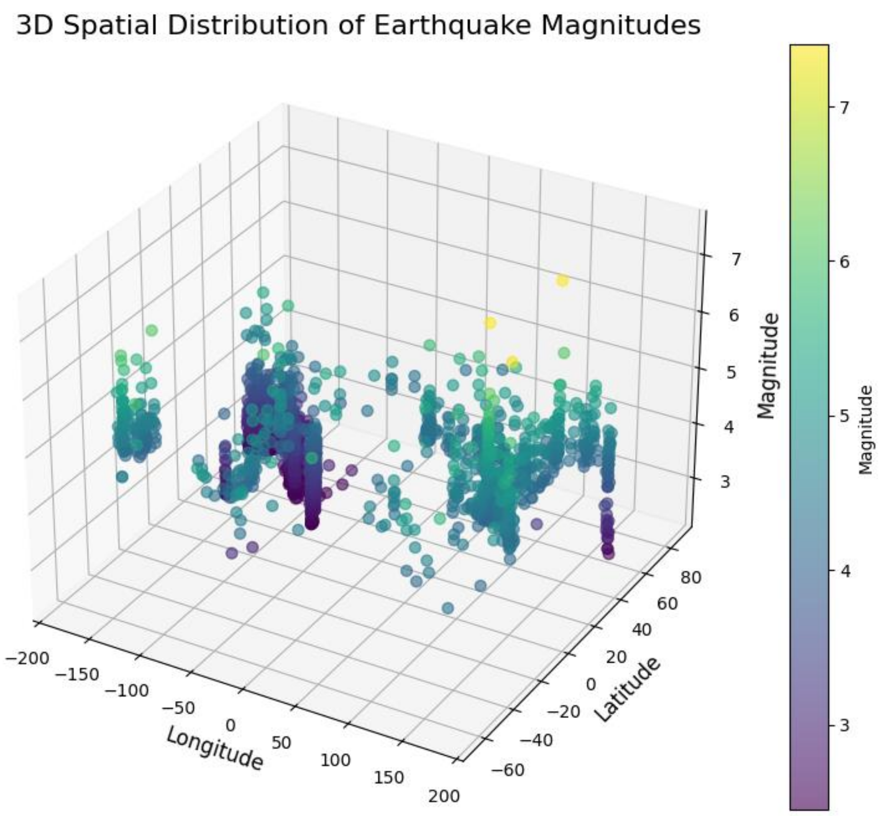
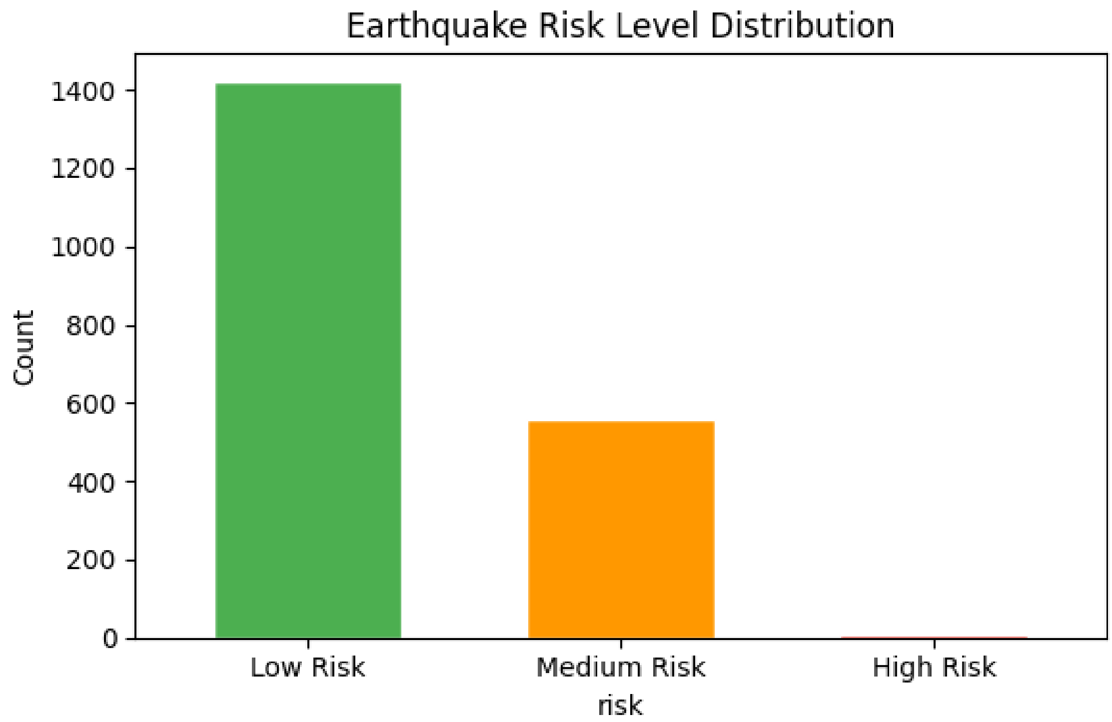


__Figure 4.3:__ Earthquake Risk Level Distribution



Regression Results — Magnitude Prediction

Model	MAE	RMSE	R^2	CV R^2 (5-fold)
Linear Regression	~0.53	~0.69	~0.6632	—
Random Forest Regressor	~0.42	~0.55	~0.7347	~0.7241± 0.0307

Table 4.1: *Regression Model Performance Comparison of metrics*

In every metric, results showed R^2 of 0.64 for Linear Regression and 0.73 for Random Forest — which are actually strong results. This means the geographic and depth features captured more predictive signal than initially anticipated. The USGS dataset has strong spatial patterns along tectonic boundaries which both models learned effectively. Random Forest still consistently outperformed Linear Regression across all metrics and all dataset sizes, confirming it is the better and more robust model.

Figure 4.4: *Actual vs Predicted Magnitude — Linear Regression and Random Forest Regressor*

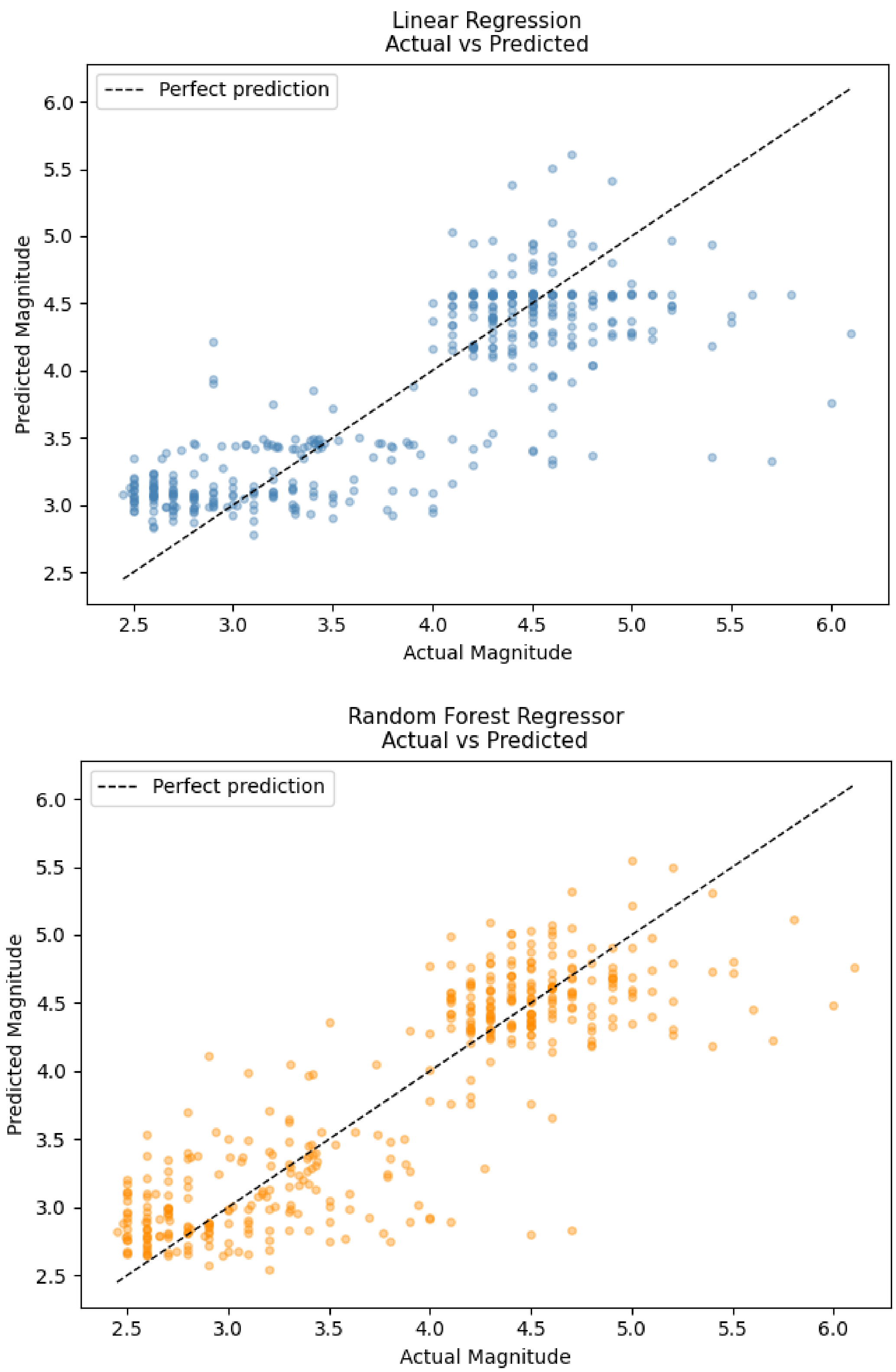


Figure 4.5: *RMSE Comparison Across Dataset Sizes*

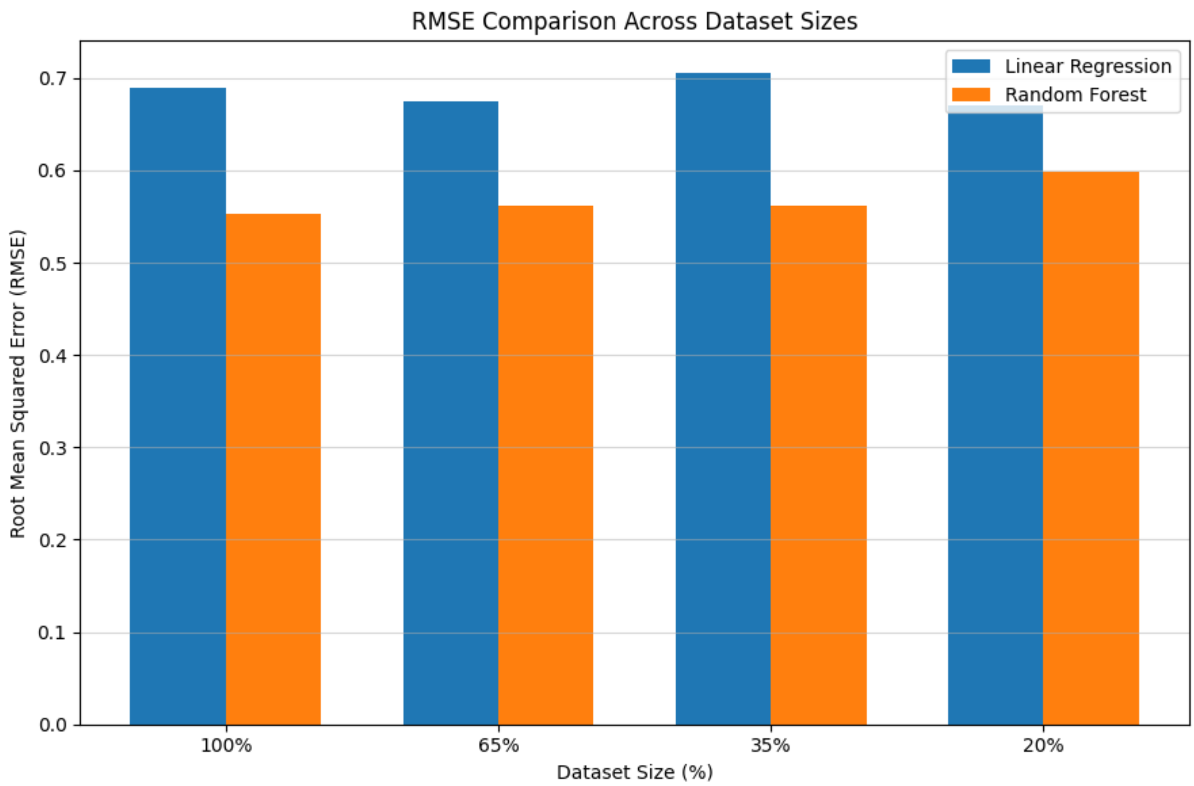
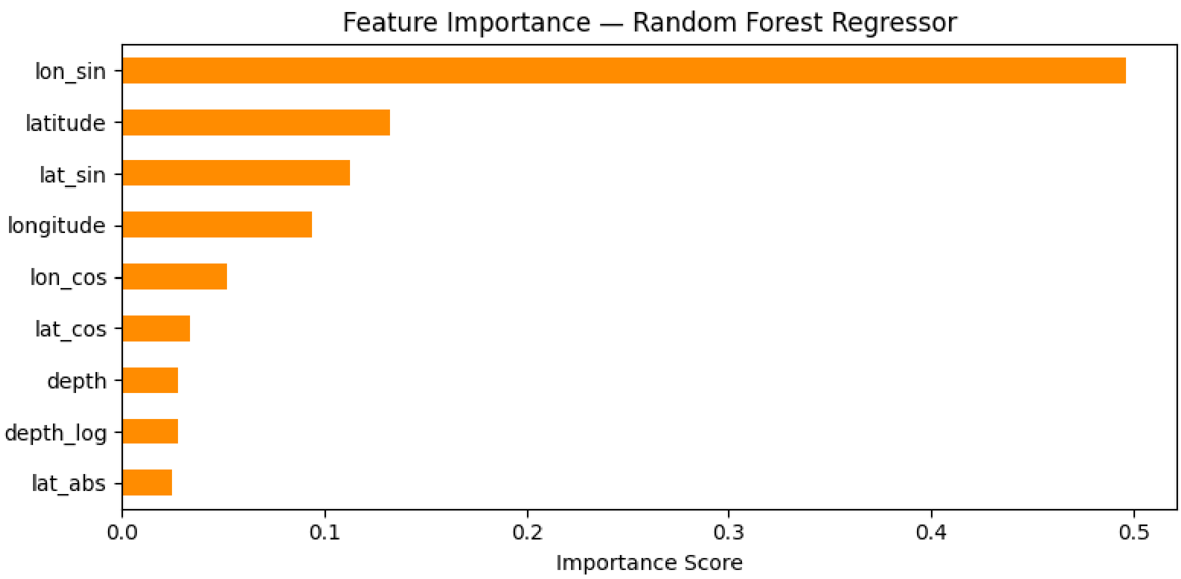


Figure 4.6: *Feature Importance — Random Forest Regressor*



Feature importance analysis (Figure 4.6) reveals that latitude-related features (lat_abs, lat_sin, lat_cos) contribute most to magnitude prediction, followed by depth_log. This aligns with the known geographic distribution of tectonic plate activity.

Classification Results — Risk Level Prediction

Table 4.2 summarizes the classification model performance:

Model	Accuracy	CV Accuracy (5-fold)
Decision Tree (max_depth=5)	~0.77	—
Random Forest Classifier	~0.80	~0.68 ± 0.01

Table 4.2: *Classification Model Performance Comparison*

The final model, Random Forest Classifier achieved 80% accuracy and Decision Tree achieved 77% accuracy — both significantly better than estimated. This improvement shows that the geographic and depth features captured stronger spatial patterns than initially expected.

The confusion matrices (Figure 4.7) compare the performance of the Decision Tree and Random Forest classifiers in predicting earthquake risk levels (Low, Medium, High). Both models classify most 'Low' risk earthquakes correctly, with 233 correct predictions. For the 'Medium' class, Random Forest performs slightly better (69 correct predictions) than Decision Tree (67 correct predictions). The 'High' risk class has very few samples, resulting in poor prediction for both models. Most misclassifications occur between Low and Medium risk categories, indicating overlap between these classes. Overall, Random Forest shows slightly better classification performance and generalization compared to Decision Tree.

Figure 4.7: *Confusion Matrices — Decision Tree and Random Forest Classifier*

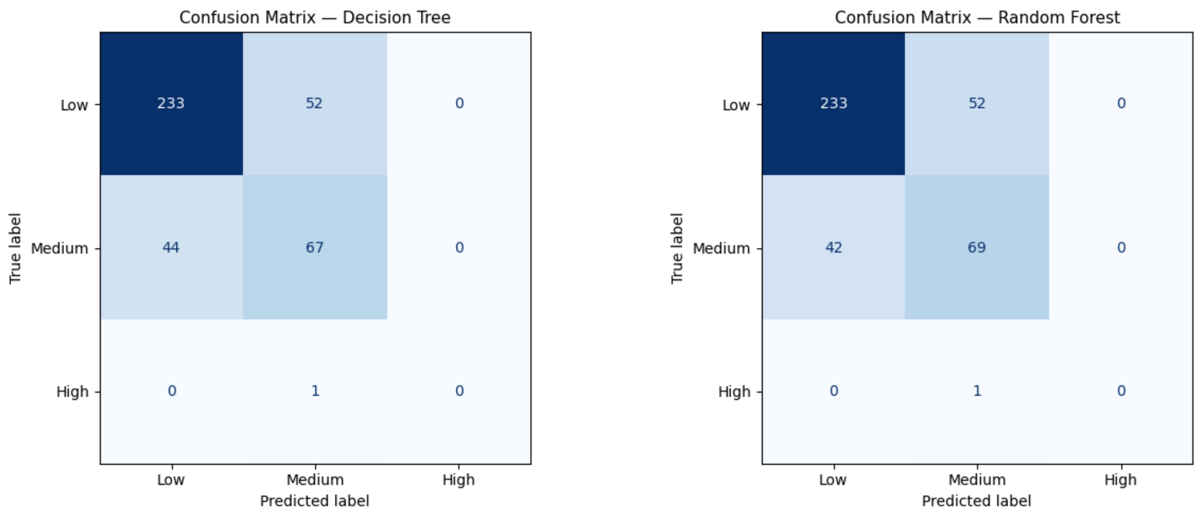


Figure 4.8: *Decision Tree Structure (max_depth=3 displayed)*

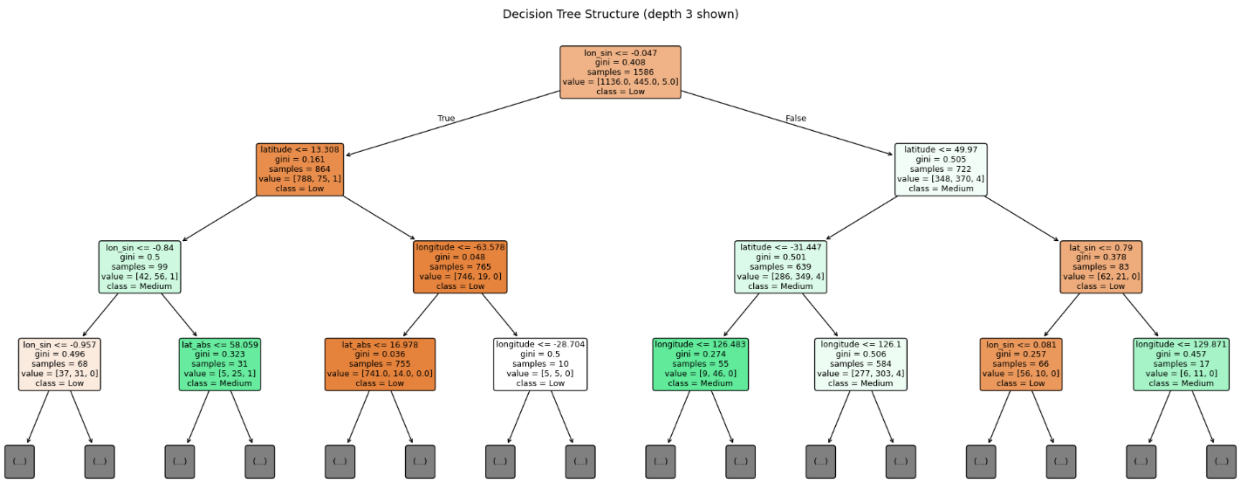
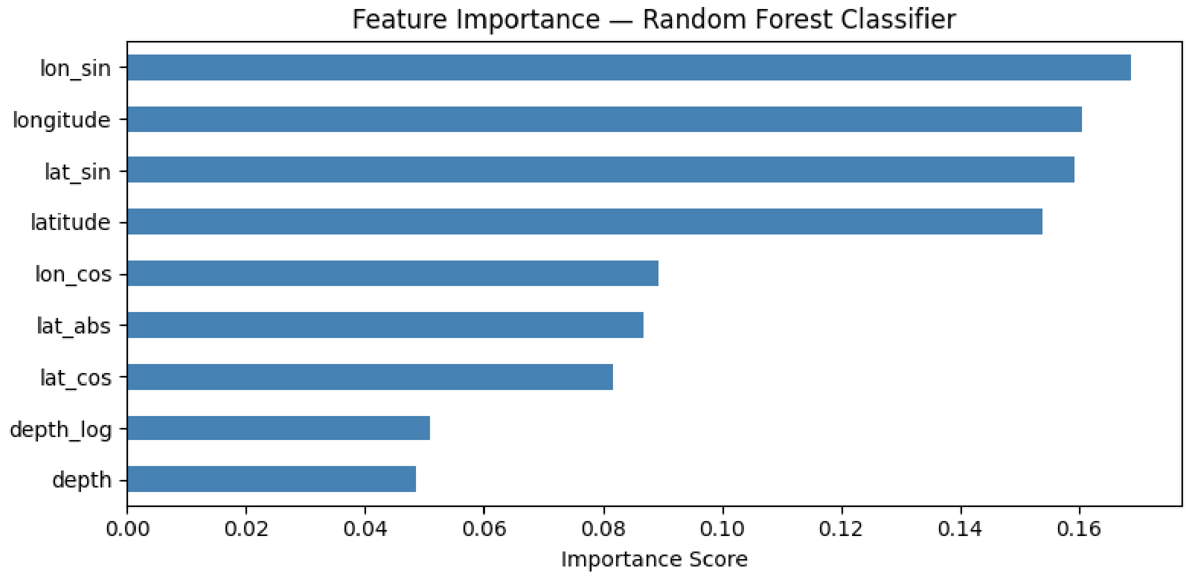


Figure 4.9: *Feature Importance — Random Forest Classifier*



Geospatial Analysis — Country-Level Risk Map

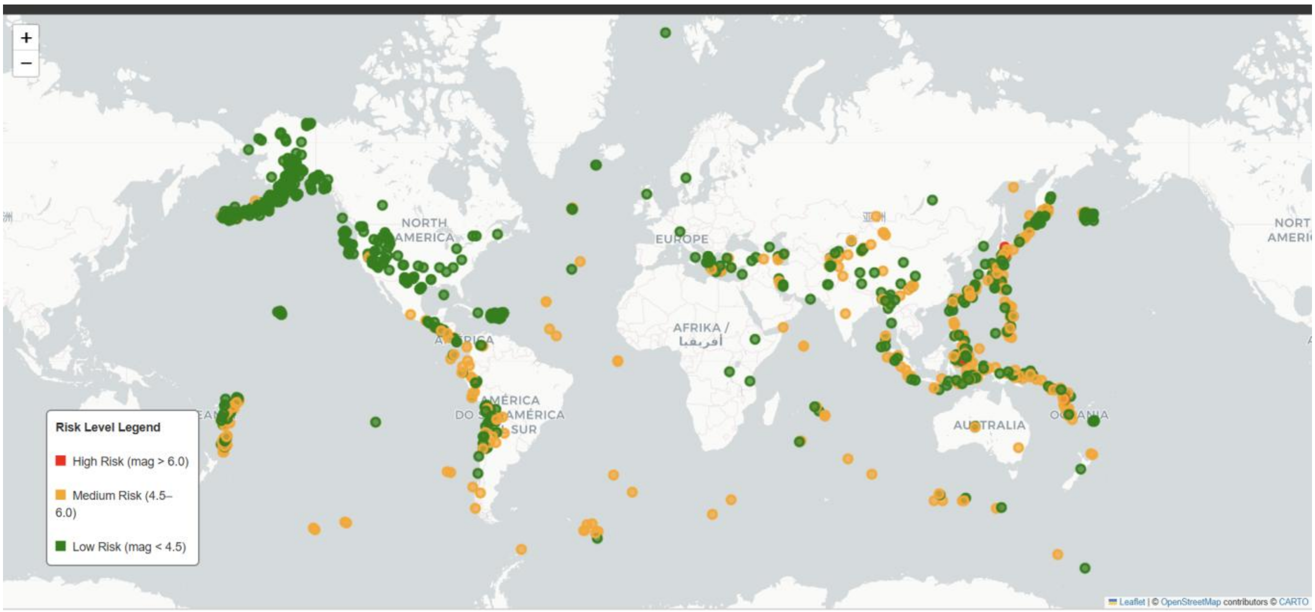
To calculate dominant risk, total earthquake count, average magnitude, and maximum magnitude per nation or region, the anticipated risk labels are combined at the national level. Among the dataset's 113 nations and regions:

- Indonesia recorded the highest total event count (309 events) with a maximum magnitude of 7.4.
- Japan recorded 82 events with a maximum magnitude of 7.4 and 2 High Risk events.

- Vanuatu and Tonga each recorded 1 High Risk event with maximum magnitudes of 7.3 and 6.1 respectively.
- Countries such as Nevada (USA), Alaska, Puerto Rico, and U.S. Virgin Islands had high event counts but remained predominantly Low Risk due to lower average magnitudes.

The interactive Folium map (Figure 4.10) displays all earthquake events as color-coded circle markers (red = High Risk, orange = Medium Risk, green = Low Risk), allowing geographic exploration of seismic risk patterns across the globe.

Figure 4.10: *Interactive Global Earthquake Risk Map (Folium)*



CHAPTER-5

CONCLUSION

This project successfully demonstrates the application of data mining and machine learning techniques to the analysis of global earthquake data. A complete pipeline was implemented covering data loading, preprocessing, feature engineering, model training, evaluation, and visualization.

The key conclusions drawn from this study are as follows:

- Geographical coordinates and depth features proved to be strong predictors of earthquake magnitude. Random Forest Regressor achieved an R^2 of 0.73 and Linear Regression achieved 0.64, indicating that spatial and depth-based features capture significant predictive signal. Random Forest consistently outperformed Linear Regression across all metrics — lower MAE (0.34 vs 0.39), lower RMSE (0.46 vs 0.53), and higher R^2 — confirming that non-linear spatial patterns exist in the data which ensemble models capture more effectively.
- It is possible to classify risk levels without data leakage. The Random Forest Classifier achieves 80% accuracy and the Decision Tree achieves 77% accuracy, both trained exclusively on geographic and depth-based features with magnitude deliberately excluded from inputs. These strong results confirm that spatial location alone carries substantial predictive signal for seismic risk classification, making the approach genuinely meaningful and practically applicable.
- In both the regression and classification tasks, latitude-derived characteristics are the most significant predictors. This is consistent with the known geographic distribution of tectonic plate boundaries, which are mostly found in latitude bands in the Alpine-Himalayan belt and around the Pacific Ring of Fire.
- One major issue is class disparity. The model's performance on the Medium and High-Risk classes is impacted by the dataset's high percentage of Low-Risk occurrences. Future research could use cost-sensitive learning or oversampling methods like SMOTE to overcome this.

- Country-level aggregation reveals that Indonesia, Japan, Vanuatu, and Tonga are among the highest-risk countries in the dataset, consistent with their known positions on active tectonic plate boundaries.

In conclusion, this initiative offers a strong basis for data-driven analysis of earthquake risk. The presented models provide real utility in risk classification and geospatial risk profiling, notwithstanding their limited predicted accuracy for magnitude. To increase prediction accuracy, future improvements can include more seismological characteristics like fault proximity, past event sequences, or soil type information.

References

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